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Google Scholar profile

Education

- 2020-2024: Ph.D. in Computer Science, University of Cyprus.
  Thesis: Hardening Modern Systems and Services for Protecting User Privacy.
  Advisor: Assoc. Prof. Elias Athanasopoulos.
- 2018-2019: M.Sc. in Computer Science with Specialization in Intelligent Systems, University of Cyprus.
  Graduated with first class honours. GPA: 9/10
  Dissertation: Assessing the Impact of Deep Learning on Internet Services' Security Mechanisms.
  Advisor: Assoc. Prof. Elias Athanasopoulos.
- 2014-2018: B.Sc. in Computer Science, University of Cyprus.
  Graduated with first class honours. GPA: 9.05/10
  Dissertation: Protein Secondary Structure Prediction using Convolutional Neural Networks in Combination with Gabor Filters and Support Vector Machines.
  Advisor: Prof. Chris Christodoulou.
- 2010-2014: CISCO CCNA Discovery.
- 2010-2012: Cambridge GCE A Level Computing.
- 2009-2012: High School Apolytirion, Kato Polemidia Lyceum, Limassol, Cyprus.
- 2010-2011: Cambridge IGCSE English as a Second Language.
- 2008-2010: Cambridge IGCSE Information and Communication Technology.
- 2007-2010: European Computer Driving License (ECDL) Certified.

Employment

- November 2026 – Present: Delft University of Technology, Assistant Professor, Cybersecurity Group.
- September 2025 – Present: Delft University of Technology, Marie Skłodowska-Curie Postdoctoral Fellow (part-time from November 2026).
- September 2024 – August 2025: Frederick University, Lecturer in Computer Security (Tenure-Track).
- September 2018 – June 2025: University of Cyprus, Research Associate at Security Research in Cyprus (SREC) group.
- July 2020 – July 2023: Resoloupe LTD, Machine Learning Engineer (part-time).
- November 2020 – March 2022: Nearix, Data Scientist/Machine Learning Engineer (contract).
- March 2016 – March 2020: University of Cyprus, Research Associate at Computational Intelligence and Neuroscience (CIN) group.
- June 2017 – August 2017: Deloitte Ltd, Software Engineer at Cyber Risk/Security Team (internship).

- July 2012 – July 2014: Soldier at Cypriot National Guard. Information and Communication Department.

Scholarships

- Huawei Technologies (Cyprus) Co. Ltd., Seeds for the Future Scholarship for Academic Excellence (2024).
- "Evagoras and Praxandros" Scholarship. University of Cyprus, Covers PhD Tuition Fees (2020-2024).
- Cyprus State Scholarship Foundation, MSc and PhD Scholarship for Academic Excellence (2018-2024).
- Logicom Public Ltd., MSc and PhD Scholarship for Academic Excellence (2018-2024).
- Graduate Research Fellowship, SREC, Dr. Elias Athanasopoulos, 2019-2024.

Awards

- Distinguished paper award finalist (IEEE EuroS&P 2022).
- 1st position at ClimateLaunchpad Climate-KIC National Finals.
- Awards of Academic Excellence for the Master student with the highest academic performance of the University of Cyprus, the Faculty of Pure and Applied Sciences and the Department of Computer Science.
- JCC Payment Systems Ltd, Award for Academic Excellence (BSc).
- iSignThis Ltd, Award for Academic Excellence in Artificial Intelligence and Systems Security (BSc).
- University of Cyprus, Department of Computer Science, Award for Academic Excellence (BSc).

Funded Research Projects

Principal Investigator

- **Personal Grant** — VALIDATE: *VALIDATing SEcurity Safeguards in Binaries Compiled with Memory-Safe Languages Pre-Execution* (1 Sep 2025 – 31 Aug 2027). **Marie Skłodowska-Curie Postdoctoral Fellowship** (Horizon Europe, MSCA PF), HORIZON-MSCA-2024-PF-01-01, Grant Agreement ID: 101206668. EU contribution: **€217,076.16**.

Research Associate

- SecOPERA: *Secure Open-Source Software and Hardware Adaptable Framework* (1 Jan 2023 – 31 Dec 2025). Horizon Europe, HORIZON-CL3-2021-CS-01.

- CyberSecPro: *Collaborative, Multi-modal and Agile Professional Cybersecurity Training Program for a Skilled Workforce in the European Digital Single Market and Industries* (1 Dec 2022 – 30 Nov 2025). Horizon Europe – Digital Europe Programme, DIGITAL-2021-SKILLS-01-SPECIALISED.

- CyberSec4Europe: *Establishing and Operating a Pilot for a Cybersecurity Competence Network to Develop and Implement a Common Cybersecurity Research and Innovation Roadmap* (1 Feb 2019 – 31 Dec 2022). Horizon 2020, H2020/SU-ICT-03-2018.

- PERSONAS: *Situational Awareness, Control and Security Policies Enforcement on Multiple Virtualization Personas of Personal Devices* (1 Oct 2018 – 30 Sep 2020). Research Promotion Foundation (ENTERPRISES/0916/0063).

Data Scientist/Machine Learning Engineer

- LightSense: *Intelligent Light Sensing for Next Generation Smart Grids* (1 Jun 2020 – 31 May 2023). Research Promotion Foundation (INTEGRATED/0918/0031).

Selected Publications

1. Jesús María Gómez Moreno, Vissarion Moutafis, Antreas Dionysiou, Fernando Kuipers, Georgios Smaragdakis, Bart Coppens, and Alexios Voulimeneas. Clair Obscure: The Light and Shadow of System Call Interposition – From Pitfalls to Solutions with K23. *In Proceedings of the 26th ACM/IFIP International Middleware Conference (MIDDLEWARE)*, December, 2025. Nashville, TN, USA.
2. Antonis Louka, Antreas Dionysiou, and Elias Athanasopoulos. Validating Memory Safety in Rust Binaries. *In Proceedings of the 17th European Workshop on Systems Security (EuroSec)*, April, 2024. Athens, Greece.
3. Antreas Dionysiou and Elias Athanasopoulos. SoK: Membership Inference is Harder than Previously Thought. *In Proceedings of the 23rd Privacy Enhancing Technologies Symposium (PETS)*, June, 2023. Lausanne, Switzerland.
4. Antreas Dionysiou, Vassilis Vassiliades, and Elias Athanasopoulos. Exploring Model Inversion Attacks in the Black-box Setting. *In Proceedings of the 23rd Privacy Enhancing Technologies Symposium (PETS)*, June, 2023. Lausanne, Switzerland.
5. Antreas Dionysiou and Elias Athanasopoulos. Lethe: Practical Data Breach Detection with Zero Persistent Secret State. *In Proceedings of the 7th IEEE European Symposium on Security and Privacy (EuroS&P)*, June, 2022. Genoa, Italy. ***Distinguished paper award finalist.***
6. Antreas Dionysiou and Elias Athanasopoulos. Unicode Evil: Evading NLP Systems Using Visual Similarities of Text Characters. *In Proceedings of the 14th ACM Workshop on Artificial Intelligence and Security (AISEC)*, November, 2021. Virtual.
7. Antreas Dionysiou, Vassilis Vassiliades, and Elias Athanasopoulos. HoneyGen: Generating Honeywords Using Representation Learning. *In Proceedings of the 16th ACM Asia Conference on Computer and Communications Security (AsiaCCS)*, June, 2021. Hong Kong, China (virtual).
8. Antreas Dionysiou and Elias Athanasopoulos. SoK: Machine vs. Machine - A Systematic Classification of Automated Machine Learning-based CAPTCHA Solvers. *In Proceedings of the Computers & Security*, 97, 101947, July, 2020.
9. Antreas Dionysiou, Michalis Agathocleous, Chris Christodoulou and Vasilis Promponas. Convolutional Neural Networks in Combination with Support Vector Machines for Complex Sequential Data Classification. *In Proceedings of the 27th International Conference on Artificial Neural Networks (ICANN)*, October, 2018. Rhodes, Greece.

Contributed Book Chapters

- Blue Book – A set of cybersecurity roadmaps and challenges for researchers and policymakers. Ch. Machine Learning, 2022. https://the-blue-book.eu/

Teaching

Instructor

- ACSC390: Software Engineering Professional Placement, Frederick University, Spring 2025.
- ACSC430: Dynamic Languages, Frederick University, Spring 2025.
- ACSC131: Internet Technologies, Frederick University, Spring 2025.
- ACOE426: Network and System Security, Frederick University, Spring 2025.
- ACSC476: Web Programming, Frederick University, Fall 2024.
- ACSC288: Data Structures, Frederick University, Fall 2024.
- ACSC478: Computational Neuroscience, Frederick University, Fall 2024.

Invited Lecturer

- CS4150: Systems Security, TU Delft, December 2025 — Invited lecture titled *"Rust Memory-Safety Checks, Cross-Language Attacks"*.

Teaching Assistant

- CS667: Computational Neuroscience, University of Cyprus, Spring 2021, 2022, 2023, 2024.
- DSC517: Data Security, University of Cyprus, Fall 2023.
- CS325: Parallel Processing, University of Cyprus, Spring 2023.
- CS444: Computational Intelligent Systems, University of Cyprus, Spring 2022, 2024.
- CS133: Object-Oriented Programming, University of Cyprus, Fall 2022.
- CS682: Advanced Security Topics, University of Cyprus, Fall 2021.

Thesis Supervision

- Cristian Comendant (MSc), *Evaluating Real-World Rust Apps for Post-Compilation Vulnerabilities*, Delft University of Technology, 2026. Supervisor.
- Dimitris Ntatsis (MSc), *Expanding Location & Validation of Modified Rust Binary Checks via Static and Dynamic Analysis*, Delft University of Technology, 2026. Supervisor.
- David Iliopoulos (MSc), *Honeywords in Practice: Industry Adoption and the Role of Secrecy & Indistinguishability in Deception-Based Defenses*, Delft University of Technology, 2026. Supervisor.
- Charalambos Thoma (BSc), *The Return of Forgotten Knowledge: On the Reversibility of Approximate Machine Unlearning*, University of Cyprus, 2026. Co-supervisor.
- Marinos Efthymiou (BSc), *Evaluating Zero-Shot LLMs for Malicious HTTP Request Detection*, Frederick University, 2025. Supervisor.
- Matthew Faghiri (BSc), *CRAWLSECCY (Crawl Security Cyprus)*, Frederick University, 2025. Supervisor.
- Panayiotis Charitonos (BSc), *Enhancing Honeywords*, University of Cyprus, 2025. Co-supervisor (MSc, KU Leuven).
- Antonis Louka (MSc), *Validating Memory Safety in Rust Binaries*, University of Cyprus, 2024. Co-supervisor (PhD candidate, KU Leuven).
- Manos Theofilou (BSc), *Clustering Crash Reports from Fuzzing Sessions using Vector Embeddings*, University of Cyprus, 2023. Co-supervisor (MSc, ETH Zürich).

Service

Program Committee

- Network and Distributed System Security (NDSS), 2027.
- IEEE Symposium on Security and Privacy (S&P), 2027.
- ACM Conference on Reproducibility and Replicability (REP), 2026.
- European Conference on Computer Systems (EuroSys), Shadow PC Member, 2026.
- Annual Conference e-Business, e-Services and e-Society (I3E), 2025.
- Annual Congress on Artificial Intelligence of Things (AIoT), 2025.
- Conference on Computer Science and Intelligence Systems (FedCSIS), 2025.
- European Workshop on Systems Security (EuroSec), 2025, 2026.
- Springer Information Security Conference (ISC), 2024, 2025, 2026.
- International Conference on Security and Cryptography (SECRYPT), 2024, 2025.
- International Conference on Security and Privacy in Communication Networks (SecureComm), 2024, 2025, 2026.
- International Conference on Network and System Security (NSS), 2024.
- Global Engineering Education Conference (EDUCON), 2025.
- International Workshop on Testing Distributed Internet of Things Systems (TDIS), 2025, 2026.
- International Conference on Advanced Information Networking and Applications (AINA), 2024.

- International Conference on Distributed Computing Systems (ICDCS), 2023, 2024.
- International Smart Cities Conference (ISC2), 2022.
- International Conference on Artificial Intelligence Applications & Innovations (AIAI), 2019.
- International Conference on Engineering Applications of Neural Networks (EANN), 2019.
- International Conference on Artificial Neural Networks (ICANN), 2018.

Reviewing for Transactions and Journals

- Elsevier Computers & Security.
- IEEE Transactions on Services Computing.
- Elsevier Frontiers of Computer Science.
- IEEE Transactions on Services Computing.
- IEEE Transactions on Dependable and Secure Computing.
- IEEE Transactions on Intelligent Systems and Technology.
- IEEE Journal of Biomedical and Health Informatics.
- Springer Neural Computing & Applications.
- Elsevier Computer Methods and Programs in Biomedicine.
- Springer Nature Scientific Reports.

Conference Organization

- Web Chair, The ACM Conference on Computer and Communications Security (CCS), 2026.
- Local Arrangements Chair, The ACM Conference on Reproducibility and Replicability (REP), 2026.

Artifact Evaluation Committee

- USENIX Security Symposium, 2020, 2024.
- International European Conference on Parallel and Distributed Computing (EuroPar), 2023.

Entrepreneurship

- September 2020 – March 2024: Ag Catalytic Solutions Ltd, Co-founder.
- March 2020 – October 2020: Climate-KIC Accelerator: Stage 2.
- February 2018 – September 2018: Ecofy Ltd, Co-founder & CEO.
- April 2018 – June 2018: Participating Startup (Ecofy Ltd) at IDEA Startup Incubator – Accelerator.
- June 2019 – September 2019: ClimateLaunchpad Climate-KIC Regional Innovation Scheme Accelerator Programme.